FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2023
FULL TEST – XI
PAPER –1
TEST DATE: 28-05-2023

ANSWERS, HINTS & SOLUTIONS

Physics

PART – I

Section – A

1. B, C

Sol. $\frac{1+1}{\gamma_{mix}-1}=\frac{1}{\frac{7}{5}-1}+\frac{1}{\frac{5}{3}-1}\Rightarrow \gamma_{min}=1.5$

$M_{mix}=\frac{1\times 4+1\times 2}{1+1}=3\text{ g/mol}=3\times 10^{-3}\text{ kg/mole}$

$v_{mix}=\sqrt{\frac{\gamma_{min}RT}{M_{imn}}}=\sqrt{\frac{1.5\times\frac{25}{3}\times 375}{3\times 10^{-3}}}=1250\text{ m/s}$

$\frac{\Delta v}{v}\times 100=\frac{1}{2}\frac{\Delta T}{T}\times 100=\frac{1}{2}\times\frac{7.5}{375}\times 100=1\%$

2. B, C

Sol. Speed of light in the medium $v=\frac{\omega}{k}=\frac{400\pi}{2\pi\times 10^{-6}}=2\times 10^{8}\text{ m/s}$

Refractive index of the medium $n=\frac{c}{V}=\frac{3\times 10^{8}}{2\times 10^{8}}=1.5$

$n=\sqrt{\varepsilon_r\mu_r}$

Since for non magnetic medium $\mu_r = 1$, $\varepsilon_r = 9/4$

Since, $B_0=\frac{E_0}{v}$ and $\vec{v}\,||\,\vec{E}\times\vec{B}$

$\vec{B}=(1.5\hat{i}+2\hat{j})\times 10^{-10}\sin(2\pi\times 10^{6}z-4\times 10^{14}\pi t)\,T$

Peak value of magnetic field = $\sqrt{(1.5)^2+(-2)^2}\times 10^{-10}=2.5\times 10^{-10}\,T$

Average energy density in electric field = $\frac{1}{2}\varepsilon_0\varepsilon_r E_{rms}^2=\frac{1}{4}\varepsilon_0\varepsilon_r E_0^2$

$=\frac{1}{4}\varepsilon_0\times\frac{9}{4}\times 25\times 10^{-4}=\frac{225}{16}\varepsilon_0\times 10^{-4}\text{ J/m}^3$

For More Material Join: @JEEAdvanced_2024

3. B, C, D

Sol. Let n_2 be the quantum number of excited state after absorption of photon.

$$\frac{n_2(n_2-1)}{2}=10 \Rightarrow n_2 = 5$$

The quantum number of energy level before excitation may be either $n_1 = 3$ or $n_1 = 2$

Let $n_1 = 2$

$$\Rightarrow \frac{256E_0}{225}=E_0z^2\left(\frac{1}{2^2}-\frac{1}{5^2}\right)$$

Here z has no integral value after calculation. So, $n_1 = 2$ is not possible

Then, $n_1 = 3$

$$\frac{256E_0}{225}=E_0z^2\left(\frac{1}{3^2}-\frac{1}{5^2}\right)$$

$$\Rightarrow \frac{256E_0}{225}=E_0z^2\frac{16}{225}$$

$\Rightarrow z = 4$

The minimum energy of the emitted photons corresponds to n = 5 to n = 4

$$\Delta E = E_0 \times 4^2\left(\frac{1}{4^2}-\frac{1}{5^2}\right)=\frac{9E_0}{25}$$

4. A, B, D

Sol. $$\frac{q_1}{q_2}=\frac{C_1}{C_2}=\frac{k\varepsilon_0\frac{A}{4d}}{\varepsilon_0\frac{A}{4d}}=\frac{k}{3}$$

since, $q_1 = q_2$ (given) $\Rightarrow k = 3$

force between the plates of a parallel plate capacitor is $F=\frac{q^2}{2A\varepsilon_0}$

$$\Rightarrow F_1=\frac{(q_/ /2)^2}{2\frac{A}{4}\varepsilon_0}=\frac{q_0^2}{2A\varepsilon_0}$$

$$F_2=\frac{(q_0/2)^2}{2\left(\frac{3A}{4}\right)\varepsilon_0}=\frac{q_0^2}{6A\varepsilon_0}$$

$$\frac{F_1}{F_2}=\frac{1/2}{1/6}=3$$

Induced charge on dielectric

$$q'=\frac{q_0}{2}\left(1-\frac{1}{k}\right)=\frac{C_0V}{2}\left(1-\frac{1}{3}\right)=\frac{C_0V}{3}$$

$$F_0=\frac{q_0^2}{2A\varepsilon_0}$$

$$F'=F_1+F_2=\frac{2q_0^2}{3A\varepsilon_0}$$

$$\frac{F'}{F_0}=\frac{2/3}{1/2}=\frac{4}{3}$$

For More Material Join: @JEEAdvanced_2024

5. B, C

Sol. $V = ar^2 - br^3$

$E = -2ar + 3br^2$

$E(4\pi r^2) = \frac{q}{\varepsilon_0} \Rightarrow -8\pi ar^3 + 12\pi br^4 = \frac{q}{\varepsilon_0}$

$\Rightarrow (-24\pi ar^2 + 48\pi br^3) = \frac{dq}{\varepsilon_0 dr} = \frac{\rho 4\pi r^2 dr}{\varepsilon_0 dr}$

$-6a + 12br = \frac{\rho}{\varepsilon_0}$

$\Rightarrow \rho = \varepsilon_0(12br - 6a) = 6\varepsilon_0(2br - a)$

$\rho = 0$

$\Rightarrow r = \frac{a}{2b}$

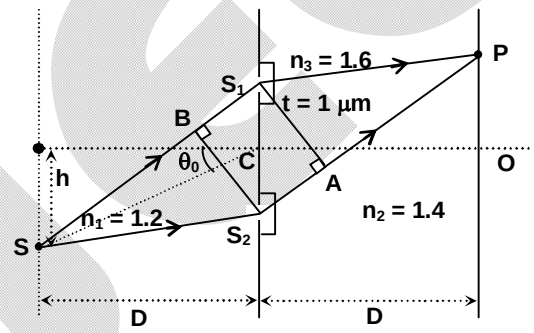
6. B, C

Sol. $\Delta x = \Delta x_1 + \Delta x_2$

$= -n_1 S_2 B + n_2 S_2 P - \{n_2(S_1 P - t) + n_3 t\}$

$= -n_1 d\frac{h}{D} + n_2(S_2 P - S_1 P) - (n_3 - n_2)t$

$= -n_1 d\frac{h}{D} + n_2\frac{dy}{D} - (n_3 - n_2)t$



(i) $h = 0$

$\Rightarrow \Delta x = n_2\frac{0}{d} - (n_3 - n_2)t = -(0.2)t = -2\times10^{-7}$ m

$\phi = -\frac{2\pi}{6\times10^{-7}}\times 2\times10^{-7} = -\frac{2\pi}{3}$

$I = I_0\cos^2\frac{\phi}{2} = \frac{I_0}{4}$

(ii) For constructive interference at P, $\Delta x = n\lambda$

$= n_2\frac{y}{D} - (n_3 - n_2)t = n\lambda$

$n_2\frac{dy}{D} = n\lambda + (n_3 - n_2)t = n\times6\times10^{-7} + 0.2\times10^{-6}$

$y = \frac{1}{10^{-3}\times1.4}\left[n\times6\times10^{-7} + 2\times10^{-7}\right] = \frac{2}{1.4}\times10^{-4} = \frac{1}{7}$ mm

(iii) At $y = 0$, $\Delta x = -x_1 d\frac{h}{D} - (n_3 - n_2)t$

$-1.2\times10^{-3}\times10^{-3} - 0.2\times10^{-6} = -1.4\times10^{-6}$

$\phi = \frac{2\pi\times1.4\times10^{-6}}{6\times10^{-7}} = \frac{14\pi}{3}$

$I = I_0/4$

(iv) $n_1\frac{d^2}{D} + n_2\frac{dy}{D} - 0.2\times10^{-6} = n\lambda$

$\Rightarrow n_2\frac{dy}{D} = n\times6\times10^{-7} + \frac{1.2\times10^{-6}}{1} + 2\times10^{-7} = n\times6\times10^{-7} + 14\times10^{-7}$

For More Material Join: @JEEAdvanced_2024

$$y = \frac{1}{10^{-3}\times 1.4}\left[-2\times 6\times 10^{-7} + 14\times 10^{-7}\right] = \frac{2\times 10^{-4}}{1.4} = \frac{1}{7}\text{ mm} \qquad (\text{for } y_{min},\ n = -2)$$

7. D

Sol. (I) frequency of sound receive by the reflector

$$f_r = \frac{v - v_r}{v} f$$

The frequency of reflected sound

$$f' = \frac{v}{v + v_r} f_r = \frac{v - v_r}{v + v_r} f = \frac{330 - 33}{330 + 33} \times 660$$

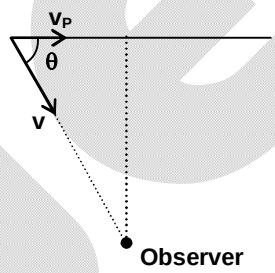
(II) frequency of sound reaching the hill

$$f_h = \frac{v}{v - v_b} f$$

The frequency of reflected sound

$$f' = \frac{v + v_b}{v} f_h = \frac{v + v_b}{v - v_b} f = \frac{330 + 16.5}{330 - 16.5} \times 627 = 693\text{ Hz}$$

(III) $f' = \dfrac{v}{v - v_P\cos\theta} f = \dfrac{v}{\left(v - v_P\dfrac{v_P}{v}\right)} f = \dfrac{v^2}{v^2 - v_P^2} f$



$$\frac{v^2}{v^2 - \left(\dfrac{v}{2}\right)^2} f = \frac{4}{3} \times 450 = 600\text{ Hz}$$

(IV) $f' = \sqrt{\dfrac{c - v}{c + v}} f = \sqrt{\dfrac{c - \dfrac{c}{2}}{c + \dfrac{c}{2}}} \times 500\sqrt{3} = 500\text{ GHz}$

8. A

Sol. (I) $\dfrac{1}{f} = (\mu - 1)\left(\dfrac{1}{R_1} - \dfrac{1}{R_2}\right)$

$$\Rightarrow \frac{1}{20} = (1.5 - 1)\left(\frac{1}{R} - \frac{1}{-R}\right) = \frac{1}{R}$$

$\Rightarrow$ R = 20 cm

For refraction at first surface (air-lens interface)

$$\frac{1.5}{v_1} - \frac{1}{-60} = \frac{1.5 - 1}{20} \qquad \ldots(i)$$

For refraction at second surface (lens-water interface)

$$\frac{4/3}{v_2} - \frac{1.5}{v_1} = \frac{\frac{4}{3} - 1.5}{-20} \qquad \ldots(ii)$$

Adding equations (i) and (ii)

$$\frac{4}{3v_2} + \frac{1}{60} = \frac{0.5}{20} + \frac{0.5}{60}$$

$\Rightarrow v_2 = 80$ cm

(II) Let F be the focal length of the silvered lens

For More Material Join: @JEEAdvanced_2024

$$\frac{1}{-f}=\frac{2}{f_e}-\frac{1}{f_m 60}=-\frac{1}{-\left(\frac{60}{2}\right)}$$

$\Rightarrow f = -15$ cm

For slivered lens

$$\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$$

$$\Rightarrow \frac{1}{v}+\frac{1}{-20}=\frac{1}{-15}$$

$\Rightarrow v = -60$ cm

(III) The system is equivalent to three thin lens in contact. Let f_2 be the focal length of the concave lens filled with water.

$$\frac{1}{f_2}=\left(\frac{4}{3}-1\right)\left(-\frac{1}{15}-\frac{1}{15}\right)$$

$$\Rightarrow f_2 = -\frac{45}{2} \text{ cm}$$

The focal length of the equivalent lens from by the three thin lenses in contact

Then, $\frac{1}{f}=\frac{1}{f_1}+\frac{1}{f_2}+\frac{1}{f_3}=\frac{1}{30}+\frac{1}{-45/2}+\frac{1}{30}$

$\Rightarrow f = 45$ cm

For the equivalent lens

$$\frac{1}{v}-\frac{1}{-60}=\frac{1}{45}$$

$\Rightarrow v = 180$ cm

(IV) For concave lens

$$\frac{1}{v_1}-\frac{1}{-30}=\frac{1}{-20} \Rightarrow v_1 = -12 \text{ cm}$$

For refraction at air convex lens interface

$$\frac{1.5}{v_2}-\frac{1}{-40}=\frac{1.5-1}{40}\Rightarrow v_2 = -120 \text{ cm}$$

For refraction at convex lens-liquid interface

$$\frac{4}{3v_3}-\frac{1.5}{-120}=\frac{\frac{4}{3}-1.5}{-40}$$

$\Rightarrow v_3 = -160$ cm

9. C

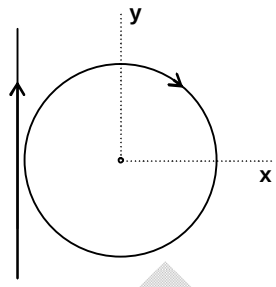
Sol. (I) $B_P=\frac{\mu_0 I}{4\pi a}+\frac{\mu_0 I}{4\pi a}=\frac{\mu_0 I}{2\pi a}$

$$B_Q=\frac{\mu_0 I}{4\pi a}(\sin 90^\circ-\sin 45^\circ)+\frac{\mu_0 I}{4\pi a}(\sin 90^\circ+\sin 45^\circ)=\frac{\mu_0 I}{2\pi a}$$

$$\frac{B_P}{B_Q}=1$$

For More Material Join: @JEEAdvanced_2024

(II) $B_P = \frac{\mu_0 \pi I}{2\pi r} + \frac{\mu_0 I}{2r} = \frac{\mu_0 I}{r}$



$B_{QZ} = \frac{\mu_0 \pi I}{2\pi\sqrt{2}r}\cos 45kd + \frac{\mu_0 I r^2}{2(r^2+r^2)^{3/2}} = \frac{\mu_0 I}{4r}\left(1+\frac{1}{\sqrt{2}}\right)$

$B_{Qx} = \frac{\mu_0 I \pi}{2\pi\sqrt{2}r}\sin 45° = \frac{\mu_0 I}{4r}$

$B_Q = \sqrt{B_{Qx}^2 + B_{Qz}^2} = \frac{\mu_0 I}{4r}\sqrt{\frac{(5+2\sqrt{2})}{2}}$

$\frac{B_P}{B_Q} = \frac{4\sqrt{2}}{\sqrt{5+2\sqrt{2}}}$

(III) $B_P = 2\times\frac{\mu_0 I}{4\pi r}(\sin 90° + \sin 45°) = \frac{\mu_0 I}{2\pi r}\left(1+\frac{1}{\sqrt{2}}\right)$

$B_Q = 2\times\frac{\mu_0 I}{4\pi r}(\sin 90° - \sin 45°) = \frac{\mu_0 I}{2\pi r}\left(1-\frac{1}{\sqrt{2}}\right)$

Here r is perpendicular distance of pints P and Q from any one straight current carrying wire

$\frac{B_P}{B_Q} = \frac{\sqrt{2}+1}{\sqrt{2}-1} = 3+2\sqrt{2}$

(IV) $B_P = 2\times\frac{\mu_0 I}{4\pi\ell}(\sin 45° + \sin 45°) = \frac{2\sqrt{2}\mu_0 I}{\pi\ell}$

$B_Q = 4\times\frac{\mu_0 I}{4\pi\frac{\ell}{\sqrt{2}}}(\sin\theta + \sin\theta)\sin 45° = \frac{\sqrt{2}\mu_0 I}{\pi\ell}\left(\frac{\ell/2}{\frac{\sqrt{3}\ell}{2}} + \frac{\ell/2}{\frac{\sqrt{3}\ell}{2}}\right)\frac{1}{\sqrt{2}} = \frac{2}{\sqrt{3}}\frac{\mu_0 I}{\pi\ell}$

$\frac{B_P}{B_Q} = \sqrt{6}$

10. C

Sol. (I) $E_P = \frac{\rho\frac{2R}{3}}{3\varepsilon_0} = \frac{2\rho R}{9\varepsilon_0}$

$E_Q = \frac{1}{4\pi\varepsilon_0}\frac{\rho\times\frac{4}{3}\pi\left(\frac{R}{3}\right)^3}{\left(\frac{2R}{3}\right)^2} = \frac{\rho R}{36\varepsilon_0}$

$\frac{E_P}{E_Q} = 8$

(II) $E_P = \frac{\rho\frac{R}{2}}{2\varepsilon_0} - \frac{1}{4\pi\varepsilon_0}\frac{\rho\times\frac{4}{3}\pi R^3\times\frac{R}{2}}{R^3} = \frac{\rho R}{12\varepsilon_0}$

$E_Q = \frac{\rho R}{2\varepsilon_0} - \frac{1}{4\pi\varepsilon_0}\frac{\rho\times\frac{4}{3}\pi R^3\times R}{R^3} = \frac{\rho R}{6\varepsilon_0}$

For More Material Join: @JEEAdvanced_2024

$$\frac{E_P}{E_Q}=\frac{1}{2}$$

(III) $B_P=\dfrac{\mu_0 i\pi R^2\times\frac{R}{2}}{2\pi R^3}=\dfrac{\mu_0 jR}{4}$

$$B_Q=\mu_0 j\frac{R}{2}-\frac{\mu_0 j\left(\frac{R}{4}\right)^2}{2\pi\left(\frac{R}{2}\right)}=\frac{7\mu_0 jR}{16}$$

$$\frac{B_P}{B_Q}=\frac{4}{7}$$

(IV) $E_P=\dfrac{\rho\times\frac{R}{2}}{2\varepsilon_0}=\dfrac{\rho R}{4\varepsilon_0}$

$$E_Q=\frac{\rho R}{2\varepsilon_0}-\frac{\rho\pi\left(\frac{R}{4}\right)^2}{2\pi\varepsilon_0\left(\frac{R}{2}\right)}=\frac{7\rho R}{16\varepsilon_0}$$

$$\frac{E_P}{E_Q}=\frac{4}{7}$$

Section – B

11. 2.24

Sol. Induced emf in loop BENM

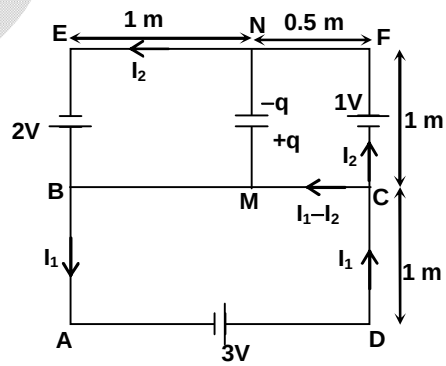
$\varepsilon_1=A_1\dfrac{dB}{dt}=1\times 2=2\text{ V}$

Induced emf in loop CFNM

$\varepsilon_2=A_2\dfrac{dB}{dt}=0.5\times 2=1\text{ V}$

Induced emf in loop ABCD

$\varepsilon_3=A_3\dfrac{dB}{dt}=1.5\times 2=3\text{ V}$



Applying loop rule in ADFE

$3-7I_1-7I_2+3=0$

$\Rightarrow I_1+I_2=\dfrac{6}{7}$ …(i)

Applying loop rule in ADCB

$3-7I_1-3(i_1-i_2)=0$

$\Rightarrow 10I_1-3I_2=3$ …(ii)

After solving equations (i) and (ii)

$I_1=I_2=\dfrac{3}{7}$

Applying loop rule in MNEF

$2-4I_2+(I_1-I_2)\times 2-\dfrac{q}{C}=0$

For More Material Join: @JEEAdvanced_2024

$$\Rightarrow \frac{q}{C} = 2 - 4 \times \frac{3}{7} = \frac{2}{7}$$

$$\Rightarrow q = \frac{2}{7} \times 7.84 = 2.24 \ \mu C$$

12. 0.11

Sol. $\Delta P_0 = |\Delta P_m \cos kx|$

$$\Rightarrow \frac{\sqrt{3}}{2}\Delta P_m = \Delta P_m \cos kx \Rightarrow kx = \frac{\pi}{6}$$

$$\Rightarrow x = \frac{\pi}{6k} = \frac{\pi}{6\frac{2\pi}{\lambda}} = \frac{\lambda}{12} = \frac{4L/3}{12} \qquad \left(\text{since } L = \frac{3\lambda}{4}\right)$$

$$= \frac{L}{9} = \frac{0.99}{9} = 0.11 m$$

13. 1.91

Sol. Q-value of reaction = $(7.01601 - 3m_e + 1.00783 - m_e) - (7.01693 - 4m_e + 1.00867) \times 931.5$ MeV
≈ -1.64 MeV

From conservation of momentum

$p_P = p_{Be}$

From conservation of energy

$$\frac{p_p^2}{2m_p} = \frac{p_{Be}^2}{2m_{Be}} + 1.64$$

$$\Rightarrow \frac{p_p^2}{2m_p}\left(1 - \frac{m_p}{m_{Be}}\right) = 1.64$$

Kinetic energy of proton = $\frac{p_p^2}{2m_p} = \frac{7}{6} \times 1.64 = 1.91$ MeV

14. 4.20

Sol. For minimum de-Broglie wavelength of electron, the momentum and hence kinetic energy of electron must be maximum

$$13.6\left(\frac{1}{1^2} - \frac{1}{2^2}\right) - \phi = \frac{h^2}{2m\lambda_1^2}$$

$$13.6z^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right) - \phi = \frac{h^2}{2m\left(\frac{\lambda_1^2}{6.1}\right)}$$

$$\frac{13.6 \times \frac{3}{4} - \phi}{13.6z^2 \times \frac{3}{4} - \phi} = \frac{1}{6.1} \qquad \ldots(i)$$

For the unknown gas

Energy of first lyman line-energy of Balmer series limit = 2×13.6 eV

$$\Rightarrow 13.6z^2\left(\frac{3}{4}\right) - 13.6z^2\left(\frac{1}{4}\right) = 2 \times 13.6$$

$z = 2$

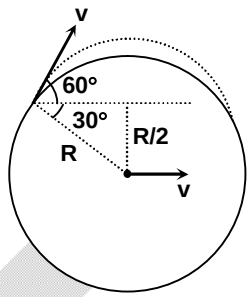
So from equation (i)

$\phi = 4.20$ eV

For More Material Join: @JEEAdvanced_2024

15. 3.46

Sol. Considering the motion of detached bit of mud w.r.t. centre of the wheel



$$2R\cos 30^\circ = \frac{v^2 \sin 120^\circ}{g}$$

$$\Rightarrow v = \sqrt{2gR} = \sqrt{2\times 10\times 0.6} = 2\sqrt{3} = 3.46\ \text{m/s}$$

16. 1.25

Sol. For block m

$(N - mg)\,\Delta t = 0 - (-mv_1) = mv_1$

For block M

$N' = Mg + N$

and $-\mu N' = (M + m)v_3 - Mv_2$

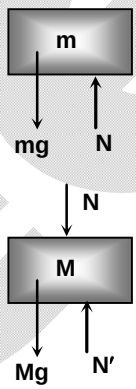
$$\Rightarrow -\mu\left(Mg + \frac{mv_1}{\Delta t} + mg\right)\Delta t = (M+m)v_2 - Mv_2$$

$Mg\Delta t$ and $mg\Delta t$ are negligible

$$\Rightarrow -\mu m v_1 = 6v_3 - 5x^2$$

$$\Rightarrow 0.25\times 1\times 10 = 6v_3 - 10$$

$$\Rightarrow v_3 = 1.25\ \text{m/s}$$

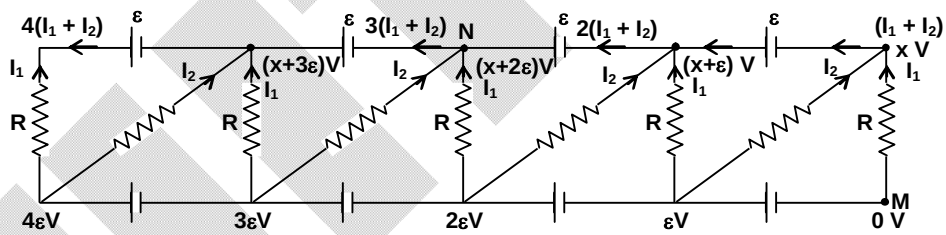


17. 5.50

Sol. Applying junction Rule at P

$$4(I_1 + I_2) + I_1 = 0 \Rightarrow I_1 = -\frac{4}{5}I_2$$

Applying loop rule in loop MTS



$$-I_1R + I_2R - \varepsilon = 0$$

$$I_2 - I_1 = \frac{\varepsilon}{R}$$

$$\Rightarrow I_2 + \frac{4}{5}I_2 = \frac{\varepsilon}{R} \Rightarrow I_2 = \frac{5\varepsilon}{9R}$$

$$\Rightarrow I_1 = -\frac{4\varepsilon}{9R}$$

$$0 - x = I_1R \Rightarrow x = -I_1R = \frac{4\varepsilon}{9}$$

$$V_N - V_M = x + 2\varepsilon - 0 = \frac{4\varepsilon}{9} + 2\varepsilon = \frac{22\varepsilon}{9}$$

$$= \frac{22}{9}\times 2.25 = 5.50\text{V}$$

For More Material Join: @JEEAdvanced_2024

18. 3.60

Sol. The work done by gas is the sum of work done against the force of atmospheric pressure and the work done against the gravity. The mercury gas interface is shifted upon the complete displacement of mercury = $\frac{2\ell}{3}+\ell+\ell=\frac{8\ell}{3}$

Work done against atmospheric pressure, $W_1=\frac{8\ell}{3}\times P_0S$

Work done against the gravity is equal to change in potential energy of mercury as a result of its displacement. The initial height of centre of mass of liquid

$$h_1=\frac{\frac{2m}{3}\times\frac{\ell}{3}+m\times 0+\frac{2m}{3}\times\frac{\ell}{3}}{\frac{7m}{3}}=\frac{4\ell}{21}$$

The final height of centre of mass of mercury = ℓ

Work done against gravity $W_2 = U_2 - U_1$

$$=\left(m+\frac{4m}{3}\right)g\left(\ell-\frac{4\ell}{21}\right)=\left(\rho s\ell+\frac{4}{3}\rho s\ell\right)g\frac{17\ell}{21}=\frac{17}{9}\rho s\ell^2 g$$

Total work done = $W_1 + W_2 = \frac{8\ell}{3}P_0S+\frac{17}{9}\rho S\ell^2 g = 3.60$ J

For More Material Join: @JEEAdvanced_2024

Chemistry

PART – II

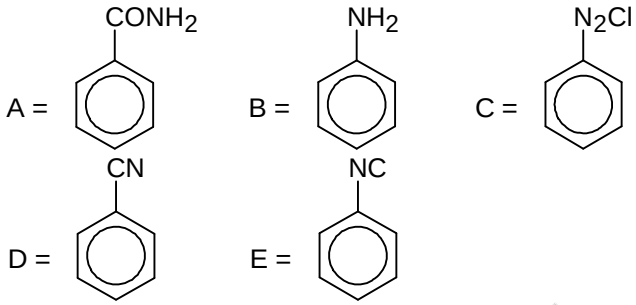
Section – A

19. A, B, D

Sol. (1) Lactose on hydrolysis produces D-Glucose and D-Galactose.
(2) Sucrose does not undergo mutarotation.
(3) D-Glucose and D-Mannose are C-2 epimers

20. B, C

Sol.



21. A, D

Sol. 1. When Chromyl Chloride vapour is passed into NaOH solution, yellow solution is formed.
2. Colour of Chromyl Chloride vapour is orange red.

22. A, B, C

Sol. Pine oil or Eucalyptus oil is added as frother.

23. A, D

Sol. 1. PH_3 when passed through $CuSO_4$ solution black ppt. (Cu_3P_2) is formed.
2. PCl_3 on hydrolysis produces H_3PO_3.

24. D

Sol. 1. Flocculation power is $PO_4^{3-} > Cl^-$.
2. Higher is the gold number, lower is the protecting power.

25. B

Sol. $[FeF_6]^{3-}$ - sp^3d^2, Paramagnetic

$[NiCl_4]^{2-}$ - sp^3, Paramagnetic

$[Ni(CN)_4]^{2-}$ - dsp^2, Diamagnetic

$[COF_6]^{3-}$ - sp^3d^2, Paramagnetic

26. D

Sol. $P_4O_{10} + 4HNO_3 \longrightarrow 4HPO_3 + 2N_2O_5$

$Pb(NO_3)_2 \xrightarrow{\Delta} 4NO_2 + 2PbO + O_2$

$2NaNO_2 + 2FeSO_4 + 3H_2SO_4 \longrightarrow Fe_2(SO_4)_3 + 2NaHSO_4 + 2H_2O + 2NO$

$NH_4NO_3 \xrightarrow{\Delta} N_2O + 2H_2O$

For More Material Join: @JEEAdvanced_2024

27. D

Sol. $2Cu(NO_3)_2 + K_4[Fe(CN)_6] \longrightarrow Cu_2[Fe(CN)_6]\downarrow + 4KNO_3$

Chocolate brown ppt.

$Pb(NO_3)_2 + K_2CrO_4 \longrightarrow PbCrO_4\downarrow + 2KNO_3$

Yellow ppt.

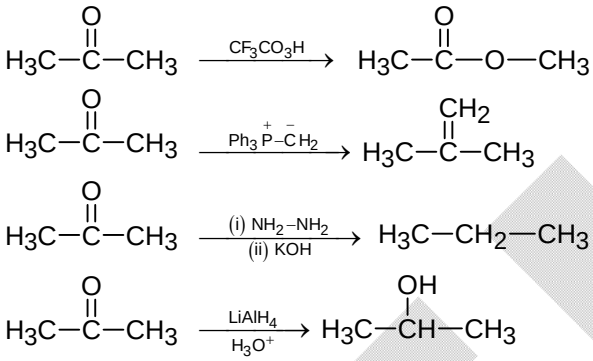
$BiCl_3 + H_2O \longrightarrow BiOCl\downarrow + 2HCl$

White ppt.

$Pb(CH_3COO)_2 + H_2S \longrightarrow PbS\downarrow + 2CH_3COOH$

Black ppt.

28. B

Sol. 

Section – B

29. 37.50

Sol. $\frac{P^0 - P^S}{P^S} = \frac{W_B / M_B}{W_A / M_A}$

$W_B = 37.50$

30. 83.50

Sol. $C_2H_6(g) + \frac{7}{2}O_2(g) \longrightarrow 2CO_2(g) + 3H_2O(\ell)$

$\Delta H^o = \left[2\Delta H_f^o\ CO_2 + 3\Delta H_f^o\ H_2O\right] - \left[\Delta H_f^o\ C_2H_6\right]$

$\Delta H_f^o\ C_2H_6 = -83.50\ kJ.$

31. 43.75

Sol. $N_2O_4(g) \rightleftharpoons 2NO_2(g)$

$\alpha = \frac{D-d}{d(n-1)} = \frac{46-32}{32(2-1)}$

$\alpha = 0.4375$

$\therefore$ % degree of dissociation = 43.75%.

For More Material Join: @JEEAdvanced_2024

32. 5.36

Sol. $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

$40 \times 0.1 \quad 32 \times 0.1 \quad 0 \quad 0$

$0.8 \quad 0 \quad 3.2$

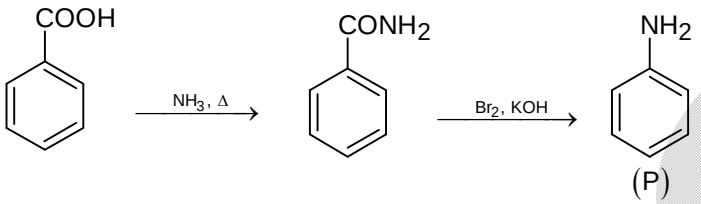
$$pH = pK_a + \log \frac{\left[CH_3COO^-\right]}{\left[CH_3COOH\right]}$$

$$pH = 4.76 + \log \frac{3.2}{0.8}$$

pH = 5.36

33. 15.05

Sol.



(P)

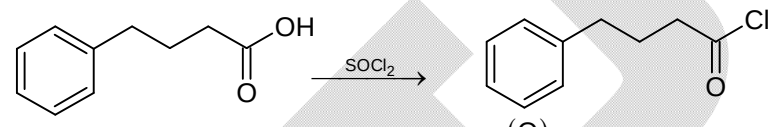
Formula of P = C_6H_7N

Molecular mass of P = 93

$$\text{Weight \% of N} = \frac{14}{93} \times 100 = 15.05$$

34. 52.80

Sol.

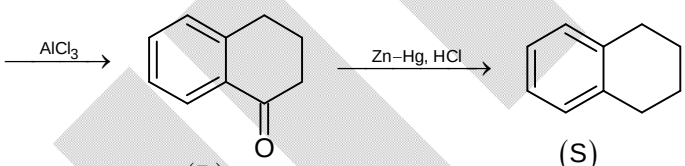


(P)

(1 mole)

(Q)

(0.5 mole)



(R)

(0,4 mole)

(S)

(0,4 mole)

Molecular weight of S = 132

$\therefore$ Wt. of compound S formed = $132 \times 0.4 = 52.8$ g

35. 2.61

Sol. $MnO_2 + 4HCl \longrightarrow MnCl_2 + 2H_2O + Cl_2$

$2KI + Cl_2 \longrightarrow 2KCl + I_2$

$2Na_2S_2O_3 + I_2 \longrightarrow Na_2S_4O_6 + 2NaI$

Meq. of MnO_2 = Meq. of $Na_2S_2O_3 = 30 \times 0.1 = 3$

$$\text{Wt. of } MnO_2 = \frac{3 \times 87}{2 \times 1000} = 0.1305 \text{ g}$$

$$\text{Wt. \% of } MnO_2 = \frac{0.1305}{5} \times 100 = 2.61$$

For More Material Join: @JEEAdvanced_2024

36. 4.32

Sol. $Ba(N_3)_2 \xrightarrow{\Delta} Ba + 3N_2$

$(0.02\ \text{mole}) \qquad (0.06\ \text{mole})$

$3Mg + N_2 \longrightarrow Mg_3N_2$

$(0.18\ \text{mole}) \quad (0.06\ \text{mole})$

Mass of $Mg = 0.18 \times 24 = 4.32\ g$

FIITJEE

For More Material Join: @JEEAdvanced_2024

Mathematics

PART – III

Section – A

37. A, B, C, D

Sol. $f'(x) = -\lambda(e^{-x} + e^{x}) \le 0$

$f(0) = 0$ and $f(x) \ge 0 \ \forall \ x \in [0, \infty)$

$\Rightarrow f(x) = 0$

$\Rightarrow \lambda = 0$

38. B

Sol. Tangent at B is perpendicular to line segment AC

39. A, B, C

Sol. $z^2 + \alpha z + \beta = 0$

$\bar{z}^2 + \bar{\alpha}\bar{z} + \bar{\beta} = 0$

$\Rightarrow \dfrac{r^4}{z^2} + \dfrac{\bar{\alpha} r^2}{z} + \bar{\beta} = 0$ (As $|z| = r$)

$\bar{\beta} z^2 + \bar{\alpha} r^2 z + r^4 = 0$

$\dfrac{1}{\bar{\beta}} = \dfrac{\alpha}{\bar{\alpha} r^2} = \dfrac{\beta}{r^4} \Rightarrow |\beta| = r^2$

40. A, B

Sol. Let $I = \int_0^1 {}^{207}C_7 x^{200}(1-x)^7 \, dx = {}^{207}C_7 \dfrac{7}{201} \int_0^1 x^{201}(1-x)^6 \, dx$

$\Rightarrow I = \dfrac{1}{208} \Rightarrow k = 13$

41. A, C, D

Sol. $k_n = 2^{n+1} - 2$

42. A, B

Sol. $r = -\dfrac{1}{4}$, $d = 2 \Rightarrow S = \dfrac{52}{25}$

43. C

Sol. Number of singular matrices with no zero entry = 209

Number of singular matrices with atleast one zero entry = 171

$\sum_{A \in S} D(A) = 10 \times (1 + 2 + 3 + ... + 9)^2 = 20250$

$\sum_{A \in S} T(A) = 100(1 + 2 + 3 + ... + 9) + 900(0 + 1 + ... + 9) = 1900 \times 45 = 85500$

$\sum_{A \in S} T'(A) = 900(1 + 2 + 3 + ... + 9) + 900(1 + 2 + ... + 9) = 1800 \times 45 = 81000$

For More Material Join: @JEEAdvanced_2024

44. A

Sol. (I) $y = e^{4x} + 2e^{-x}$

$\Rightarrow \frac{dy}{dx} = 4e^{4x} - 2e^{-x}, \frac{d^2y}{dx^2} = 16e^{4x} + 2e^{-x}$

$\frac{d^3y}{dx^3} = 64e^{4x} - 2e^{-x} \Rightarrow k = 12$

(II) Since the degree of the equation is 2, so two lines are possible

(III) $y = t^m \Rightarrow \frac{dy}{dx} = mt^{m-1} \cdot \frac{dt}{dx}$

$\Rightarrow \frac{dt}{dx} = \frac{4x^6 - t^{4m}}{2mx^4t^{2m-1}} \Rightarrow m = \frac{3}{2}$

(IV) $y = Ax^m + Bx^{-n} \Rightarrow \frac{dy}{dx} = Amx^{m-1} - Bnx^{-n-1}$

$\Rightarrow m = 3$ or -4 and $n = 4$ or -3

45. C

Sol. Let E_i denote the event that the bag contains i black and (12 – i) white balls (i = 0, 1, 2, ..., 12) and A denote the event that the four balls drawn are all black. B denote the probability of drawing 2 black and 2 white balls

$P(E_i) = \frac{1}{13}(i = 0, 1, 2, ..., 12)$, $P\left(\frac{A}{E_i}\right) = 0$ for i = 0, 1, 2, 3

$P\left(\frac{A}{E_i}\right) = \frac{{}^{i}C_4}{{}^{12}C_4}$ for $i \geq 4$

(I) $P(A) = \sum_{i=0}^{12} P(E_i).P\left(\frac{A}{E_i}\right) = \frac{1}{5}$

(II) $P\left(\frac{A}{E_{10}}\right) = \frac{{}^{10}C_4}{{}^{12}C_4} = \frac{14}{33}$

(III) $P\left(\frac{E_{10}}{A}\right) = \frac{\frac{1}{13} \cdot \frac{14}{33}}{\frac{1}{5}} = \frac{70}{429}$

(IV) $P(B) = \frac{1}{5}$

46. D

Sol. (I) $p^2 + 9q^2 + 25r^2 = pqr\left(\frac{15}{p} + \frac{5}{q} + \frac{3}{r}\right) \Rightarrow p = 3q = 5r$

(II) $(r - p)^2 + (q + r)(p - 2q + r) \Rightarrow \frac{2}{q} = \frac{1}{p} + \frac{1}{r}$

(III) $f(x) = px^2 + qx + r \Rightarrow f'(x) = 2px + q$

$f(1) = f(-1) \Rightarrow q = 0$

Also, $2p^2 + 2pr = 0 \Rightarrow$ p, q, r are in A.P.

(IV) p = 5, q = 8, r = 12

For More Material Join: @JEEAdvanced_2024

Section – B

47. 2.00

Sol. Let P and Q be $\left(at_1^2, 2at_1\right)$ and $\left(at_2^2, 2at_2\right)$ respectively $t_2 = -t_1 - \frac{2}{t_1}$

$\Rightarrow t_1(t_1 + t_2) = -2$

$\tan\alpha = -t_1$, $\tan\beta = -t_2$

$\Rightarrow |\tan\alpha(\tan\alpha + \tan\beta)| = 2$

48. 1.00

Sol. θ_1 and θ_0 are the roots of the equation $\frac{\cos\theta}{\cos\theta_2} + \frac{\sin\theta}{\sin\theta_2} = 1 \Rightarrow \frac{\cos\theta}{\cos\theta_2} = 1 - \frac{\sin\theta}{\sin\theta_2}$

$\Rightarrow \frac{\cos^2\theta}{\cos^2\theta_2} = 1 + \frac{\sin^2\theta}{\sin^2\theta_2} - 2\frac{\sin\theta}{\sin\theta_2} = \frac{1-\sin^2\theta}{\cos^2\theta_2} \Rightarrow \frac{\sin\theta_0\sin\theta_1}{\sin^2\theta_2} = -\sin^2\theta_2$

Similarly, $\frac{\cos\theta_0\cos\theta_1}{\cos^2\theta_2} = -\cos^2\theta_2 \Rightarrow \left|\frac{\sin\theta_0\sin\theta_1}{\sin^2\theta_2} + \frac{\cos\theta_0\cos\theta_1}{\cos^2\theta_2}\right| = 1$

49. 9.00

Sol. $S = \sum_{n=1}^{9999} \frac{1}{\left(n^{1/2} + (n+1)^{1/2}\right)\left(n^{1/4} + (n+1)^{1/4}\right)} = \sum_{n=1}^{9999}\left((n+1)^{1/4} - n^{1/4}\right)$

$\Rightarrow S = (10000)^{1/4} - 1^{1/4} = 9$

50. 3.00

Sol. Let $(x + 3) = -t \Rightarrow 3\sqrt{2t} - 1 = t + k$

$\Rightarrow 18t = t^2 + 2(k + 1)t + (k + 1)^2$

As $t > 0 \Rightarrow k < \frac{7}{2} \Rightarrow [k] \le 3$

51. 4.00

Sol. Let $I = \int_0^1 \frac{\sin^{-1}\left(\sqrt{x}\right)}{x^2 - x + 1}dx \Rightarrow I = \int_0^1 \frac{\sin^{-1}\left(\sqrt{1-x}\right)}{x^2 - x + 1}dx = \int_0^1 \frac{\cos^{-1}\left(\sqrt{x}\right)}{x^2 - x + 1}dx$

$\Rightarrow 2I = \frac{\pi}{2}\int_0^1 \frac{dx}{x^2 - x + 1} = \frac{\pi^2}{3\sqrt{3}} \Rightarrow I = \frac{\pi^2}{6\sqrt{3}} \Rightarrow n = 4$

52. 3.00

Sol. $\cos^{-1}(2x) + \cos^{-1}(3x) = \pi - \cos^{-1}(x) \Rightarrow 6x^2 - \sqrt{1-4x^2}\sqrt{1-9x^2} = -x$

$\Rightarrow 12x^3 + 14x^2 - 1 = 0 \Rightarrow a = 12, b = 14, c = 0, d = -1$

53. 9.00

Sol. By using $(1+x)^n = 1 + nx + \frac{n(n-1)}{2}x^2 + ...$

$L = \frac{3}{4}$

54. 49.00

Sol. $\sum_{r=1}^{49} \frac{\left(2r^2 - 48r + 1\right)}{(50-r)\,{}^{50}C_r} = \sum_{r=1}^{49}\left(\frac{r+1}{{}^{50}C_{r+1}} - \frac{r}{{}^{50}C_r}\right)$

For More Material Join: @JEEAdvanced_2024